

CLAIRE SHEN

Active DoD Secret Clearance

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Artificial Intelligence student at UC San Diego who excels at distilling user feedback into actionable plans, leading teams to implement technical solutions, and delivering products that solve operational pain points.

EDUCATION

University of California, San Diego — Jacobs School of Engineering San Diego, CA
B.S. Artificial Intelligence Expected June 2029
• **Relevant Coursework:** Data Structures, Object-Oriented Programming, Intro to AI, Linear Algebra, Data Science

EXPERIENCE

Naval Surface Warfare Center (NSWC) Corona | Range Systems Engineering Norco, CA
AI/ML Engineering Intern (NREIP) June 2026 – Aug 2026

- Built a locally run video annotation application to label sparse target events buried in hours of video
- Placed 3rd of 30 interns for the final technical presentation to NSWC Corona base leadership
- Scoped out and designed a PDF ingestion pipeline for a database of several hundred unstructured failure reports during a 3-week sprint; Implemented layout-aware PDF to text conversion using Docling before project handoff

NSWC Corona Fallbrook Detachment | Rapid Prototyping Lab Fallbrook, CA
Engineering Intern (SEAP) June 2025 – Aug 2025

- Set up daily standups and shared workspaces among 4 intern teammates to accelerate development of an AI-enabled vehicle damage assessment device; Initiated regular communications with the supervising engineer
- Collected, cleaned, and labeled a dataset of +1,000 vehicle corrosion images; Fine-tuned and deployed YOLO image segmentation model on Raspberry Pi and Google Coral Edge TPU

Engineering Intern (SEAP) June 2024 – Aug 2024

- Designed, manufactured, and tested a physical part based on loose product requirements from a machinist. Coordinated directly with Marine end users to deploy on base.
- Reverse-engineered and 3D printed hard-to-source vehicle parts with Fusion360 and SOLIDWORKS

PROJECTS

Desktop Video Annotation Application | NREIP Internship July 2026 – Aug 2026

- Gathered requirements from engineer users and built a full-stack (Python Qt, OpenCV) desktop labeling application
- Delivered video scrubbing, event-based frame extraction, and visual annotation features
- Fine-tuned and integrated YOLO26 for label proposals, cutting weekly labeling from 8 hours to 2 across 100GB of video
- Drove development with Claude Code, going from user interviews to production in a 4-week sprint

HouseMeds | Household Prescription Medication Cost Optimizer Sept 2026 – Present

- Interviewed real patients and assembled 5 person team to develop a full-stack web application that finds and recommends the lowest prices for a household's prescription medications
- Implementing Typescript and PostgreSQL backend with REST API calls to online pharmacies and the price optimization algorithm

DermLens.ai | Automated Monitoring for Skin Cancer Treatment Sept 2023 – Present

- Assisted dermatology clinic staff in collecting 5,000+ longitudinal clinical images of skin cancer treatment for a custom dataset
- Fine-tuned and evaluated CNN models for binary classification of erythema (redness)

ACTIVITIES & LEADERSHIP

Affiliations: Women in Computing (WiC); Society of Women Engineers (SWE); Women in Defense (WID)

Founder & President | SWENext Club of Murrieta Valley High School Aug 2023 – June 2025

- Founded the chapter and grew it to 40+ students with monthly technical speakers and a mentorship pipeline

Co-Captain | VEX Robotics Team of Murrieta Valley High School Aug 2022 – June 2025

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, Javascript/HTML/CSS, Typescript

Software Engineering: Object-Oriented Programming, Data Structures, REST APIs, Git, Docker, Unit Testing, Agentic Coding

AI/ML: PyTorch, TensorFlow, OpenCV, pandas, scikit-learn, Computer Vision, LLM Orchestration, RAG

Embedded & Edge: Edge Model Deployment, Google Coral Edge TPU, Raspberry Pi

Engineering: SOLIDWORKS, Fusion 360, Finite Element Analysis Simulation, Reverse Engineering, Additive Manufacturing, 3D Scanning